

WORK PROGRESS UPDATE

FocusAInt (study-streak / focusaint)

Reporting Period: March 1, 2026 – April 15, 2026

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Branch: main

About the Project

FocusAInt is an anti-distraction learning platform designed to solve the online course completion crisis. Rather than offering unlimited content, the platform enforces structured learning through daily video limits, AI-generated comprehension quizzes, mandatory reflections, and a streak-based engagement system. The product is built to serve students, career switchers, and competitive exam aspirants who struggle to finish what they start.

The platform is built on a modern stack: Next.js on the frontend, Redis for caching, MongoDB as primary database, and the Gemma for AI quiz generation. Payments are handled via Dodo Payments, and the backend is deployed on AWS.

Progress at a Glance

Feature / Area	Status
Quick Mode (Deep Focus Session)	✓ Completed
Payment Gateway Integration (Dodo Payments)	✓ Completed
Rate Limiting & Server-Side Caching	✓ Completed
Redis Setup & Stability Fixes	✓ Completed
MongoDB / Mongoose Connection Stability	✓ Completed
Backend Routing & AWS Deployment Config	✓ Completed
Landing Page Redesign (Version 2)	✓ Completed
Dashboard UI	✓ Completed
Goals Tracker Feature	✓ Completed
Deep Mode (Immersive Learning Flow)	🔄 In Progress
Habit Planner Module	🔄 In Progress

Week-by-Week Activity Log

Week 1 — Mar 01 to Mar 07, 2026

- Quick Mode baseline implemented — the core focused-session flow allowing users to enter a rapid-fire learning session without navigation distractions.
- Build error in the reset-password flow was identified and removed, restoring clean production builds on the main branch.

Week 2 & 3 — Mar 08 to Mar 21, 2026

No commits were recorded on the main branch during this period. Work was likely ongoing in feature branches or focused on planning and design reviews for the upcoming infrastructure sprint.

Week 4 — Mar 22 to Mar 28, 2026

- Rate limiting added to the backend API to prevent abuse and protect server resources as the user base grows.
- Server-side caching layer implemented to reduce redundant API calls, particularly for quiz generation and video metadata fetches — directly reducing Claude API costs per session.
- Payment gateway integrated using Dodo Payments. This enables the platform to accept subscription payments from Indian users, unlocking the freemium-to-paid conversion funnel. The integration covers order creation, payment verification, and subscription tier activation.
- .gitignore updated to exclude sensitive config files and environment variables from the repository, improving security hygiene.

Week 5 — Mar 29 to Apr 04, 2026

This was the heaviest infrastructure week of the reporting period. Significant effort went into stabilising the backend so it can reliably support concurrent users.

Deployment & Configuration

- Redis configuration updated to align with the production environment on AWS.
- backend config revised to correctly route serverless function requests.
- Trust proxy setting enabled on the Express/Node server — required for correct IP detection behind AWS's EC2.
- Environment variable configuration cleaned up to remove conflicts between staging and production environments.

Database & Cache Stability

- Redis initialization error resolved — the cache layer was failing silently on cold starts, which was fixed to ensure consistent caching on every request.
- Redis URL parsing error fixed — malformed connection strings were causing intermittent failures when the server spun up.
- Three separate MongoDB and Mongoose connection issues were identified and resolved across multiple commits. These covered initial connection setup, connection pooling errors, and a reconnection bug that caused the app to fail after periods of inactivity.

Routing

- Root route (/) correctly configured — the application now serves the landing page reliably on the primary domain.
- Default fallback route set up to handle unrecognized paths gracefully without crashing the server.

Week 6 & 7 — Apr 05 to Apr 15, 2026

No commits were pushed to the main branch during this window. Ongoing work during this period involved UI development and feature work in parallel branches, covered in the section below.

New Features Built This Period

Payment Gateway — Dodo Payments Integration	
What was built	Full Dodo Payments payment flow including order creation, payment capture, and subscription tier activation on successful payment.
Why it matters	Enables the freemium-to-paid conversion that the business model depends on. Indian market payments are now fully supported.
Pricing tiers	Free (2 videos/day), Pro at ₹99/month (5 videos/day + unlimited AI quizzes + full spaced repetition), Annual at ₹999/year.
Status	Live on main branch. Checkout and verification endpoints operational.

Landing Page — Complete Redesign (Version 2)	
What was built	A completely redesigned Version 2 landing page — rebuilt from scratch with a new visual identity, improved conversion flow, and clearer communication of the platform's value proposition: structured, distraction-free learning with AI-powered comprehension testing.
Key sections	Hero with instant call-to-action, problem statement (the 95% dropout crisis), how it works (5-step flow), social proof, and pricing tiers.
Design direction	Clean, minimal layout with strong typography. Emphasis on constraint as a feature, not a limitation.
Status	Completed and merged.

Dashboard UI — Overhaul	
What was built	Rebuilt dashboard showing the user's active learning path, daily progress ring, streak counter, recent quiz scores, and upcoming spaced-repetition reviews.
Key components	Completion ring (watch-time vs. daily limit), streak display with freeze indicator, leaderboard widget showing user's weekly rank, and the skill-tree progress visualization for their active learning path.
Status	Completed and merged.

Goals Tracker	
What was built	A dedicated goals module where users define what they want to master, set a target duration (e.g., 30-day path), and track completion day-by-day.
How it works	User sets a goal (e.g., 'Learn React Hooks'). The AI generates a structured 30-day learning path. The tracker shows completed days, current day, and estimated finish date based on daily consistency.
Data stored	Goal name, target topic, learning path ID, current day, completion status, and timestamps — persisted in Supabase.
Status	Completed and merged.

Quick Mode	
What was built	A focused session mode that strips away the full dashboard UI and drops the user directly into a video-quiz-summary loop without any navigation.
Intended use	Users who want to complete one focused learning block without going through the full learning path interface. Ideal for mobile and quick study breaks.
Status	Core baseline merged in Week 1 (Mar 3). Refinements ongoing in feature branch.

Deep Mode (In Progress)	
What is being built	An immersive, distraction-eliminating learning experience. Full-screen video player, website blocker active during playback, tab-switch detection that auto-pauses the video, and accountability partner notifications.
Current state	Core architecture defined. Full-screen player and tab-detection logic are being built. Website blocking integration is planned for next sprint.
Status	In progress — expected to complete in the next two-week sprint.

Technical Health & Infrastructure

The most important outcome of this reporting period is that the backend is now stable. All major database and cache issues that were blocking reliable operation have been resolved. Here is a summary of the current infrastructure state:

Infrastructure Status	
Frontend	Next.js 14 + React, deployed on AWS Lambda. Mobile-responsive PWA. Auto-deploys from main branch.
Database (Primary)	Mongodb. Real-time subscriptions active for streak and leaderboard updates.
Cache Layer	Redis now stable. Initialization and URL parsing errors resolved. Used for quiz caching and session data.

AI / Quiz Engine	Gemma 4 integrated. Quizzes are generated from video transcripts and cached per video+difficulty combination to control API costs.
Payments	Dodo Payments fully integrated. Order creation and payment verification endpoints live.
Serverless Functions	AWS Lambda used for quiz generation, transcript extraction, spaced-repetition scheduling, and notification dispatch.
Deployment	Trust proxy configured, environment variables cleaned up, routing stable on AWS.

Challenges Encountered

- Redis and MongoDB connection errors consumed significant time in Week 5. Multiple root causes were found across cold starts, URL parsing, and reconnection logic — each required a separate fix.
- Two weeks (Weeks 2–3 and Weeks 6–7) showed no main-branch commits. Some of this reflects work in feature branches, but it also highlights the need for more regular merges to keep the incubation timeline accurate.
- Deep Mode is running slightly behind schedule due to the infrastructure stabilisation work taking priority in late March.

What Comes Next

- Complete and merge Deep Mode — full-screen player, tab-detection, website blocker, and accountability partner notifications.
- Complete Habit Planner module — daily learning habit scheduling, reminder configuration, and streak integration.
- Beta testing with 50 users — collect feedback on the full learning loop (video → quiz → summary → unlock → spaced review).
- Product Hunt launch preparation — landing page, demo video, and launch copy.
- Target metrics for Month 2: 500 registered users, 30% Day-7 retention rate.
- Target metrics for Month 3: 1,000 users, 50 paying subscribers, ₹5,000 MRR.

Summary Note to Incubation Center

The past two months have covered the full range from core feature building to infrastructure stabilisation. The Dodo Payments integration, completely redesigned Version 2 landing page, dashboard overhaul, goals tracker, and Quick Mode together represent a significant step toward a launch-ready product. The backend is now in a stable state after resolving all critical Redis and MongoDB issues, giving a reliable foundation for the upcoming beta launch.

The team is on track to hit the Month 2 beta targets. Deep Mode and Habit Planner — the two remaining major features — are actively in development and will be completed in the next sprint. Feedback from the incubation center on any specific areas to prioritise is welcome.

Submitted by the FocusAlnt Team